

MECHATRONICS SYSTEM INTEGRATION (MCTA 3203)

SEMESTER 1 2025/2026

WEEK 7: PLC INTERFACING WITH MICROCONTROLLER AND PC OVER
ETHERNET/IP

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SECTION 1

GROUP 13

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ABSTRACT

This experiment focuses on integrating a Programmable Logic Controller (PLC) with a microcontroller and a personal computer using Ethernet/IP communication, demonstrating a modern approach to industrial automation and control. Using the OpenPLC Editor, ladder logic programs were created, simulated, and deployed onto an Arduino microcontroller to operate basic control circuits such as a blinking LED system and a Start–Stop control circuit. The exercise introduces students to essential concepts in PLC programming, variable configuration, physical pin mapping, and hardware interfacing with external components. By completing the simulation and hardware implementation, this experiment enhances understanding of PLC–microcontroller interoperability, real-time process control, and the application of IEC 61131-3 programming languages in practical mechatronics systems.

TABLE OF CONTENTS

ABSTRACT	2
TABLE OF CONTENTS	2
INTRODUCTION	3
MATERIALS AND EQUIPMENT	3
EXPERIMENTAL SETUP	4
METHODOLOGY	5
DATA COLLECTION	9
DATA ANALYSIS	9
RESULT	11
DISCUSSION	12
CONCLUSION	13
RECOMMENDATIONS	14
APPENDICES	15
ACKNOWLEDGEMENTS	15
STUDENTS DECLARATION	16

INTRODUCTION

Programmable Logic Controllers (PLCs) play a crucial role in industrial automation, where they are used to control electromechanical systems in manufacturing plants, machinery, and automated processes. With the increasing need for flexible and low-cost automation solutions, the ability to interface PLC systems with microcontrollers and personal computers has become an essential skill for engineers. Ethernet/IP communication enables seamless data exchange between devices, providing a foundation for Industry 4.0 and interconnected smart systems.

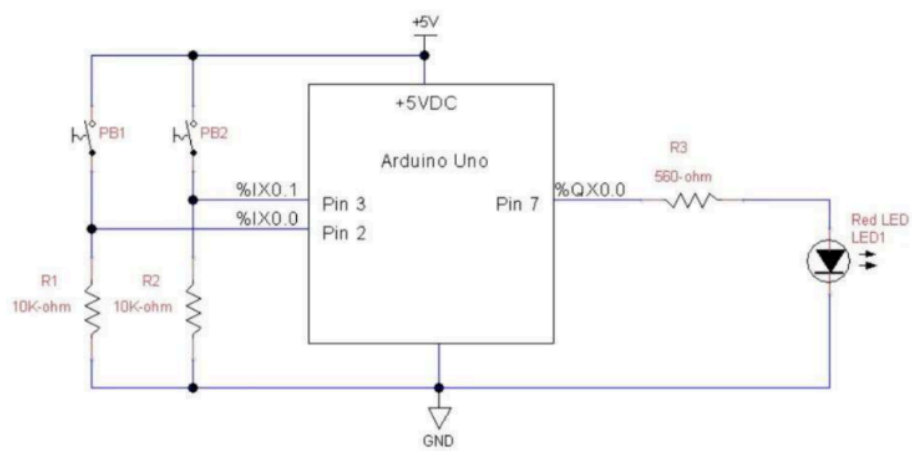
In this experiment, the OpenPLC platform is used as an open-source alternative to conventional industrial PLCs. The OpenPLC Editor allows users to design ladder logic programs in an IEC 61131-3 standard language, simulate their operation, and upload them to microcontroller-based hardware such as the Arduino. Through this platform, students learn how to create ladder diagrams, define variables, associate program elements with physical microcontroller pins, and implement real-time control tasks.

The practical sessions include developing a blinking LED ladder logic program and constructing a Start–Stop control circuit using push buttons and an LED. These exercises reinforce fundamental concepts in digital I/O control, timer usage, and PLC logic sequencing. By integrating both software simulation and physical hardware implementation, this experiment provides hands-on exposure to PLC–microcontroller interfacing, preparing students for real-world mechatronics and automation applications.

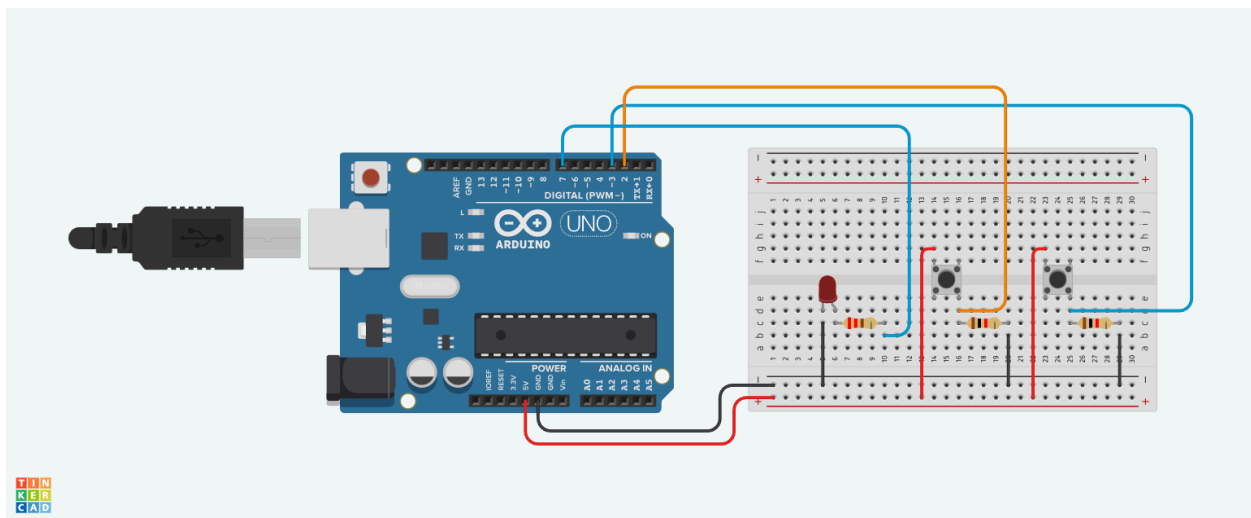
MATERIALS AND EQUIPMENT

- OpenPLC Editor software
- Arduino Board
- 2 Push Button Switches
- LED
- Resistors
- Jumper Wires
- Breadboard

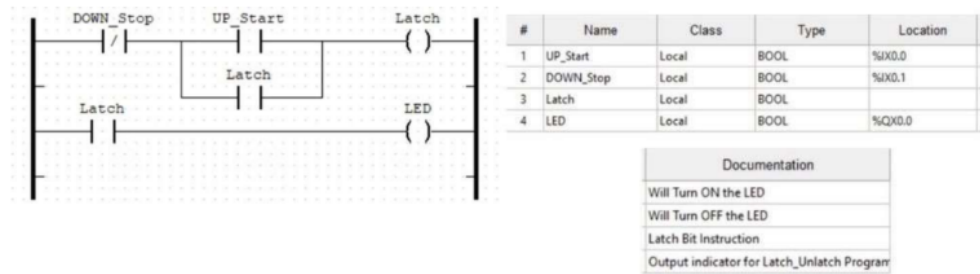
EXPERIMENTAL SETUP



Circuit Setup



Circuit Design



PLC Ladder Diagram

METHODOLOGY

1. Create a new ladder diagram in OpenPLC following the design of the module.
2. Define variables for:
 - Start button input
 - Stop button input
 - Coil (output latch)
3. Implement a latching circuit using:
 - NO (normally open) contacts
 - NC (normally closed) contacts
 - Output coil
4. Simulate the logic to ensure the Start button latches the output and the Stop button resets it.
5. Assign physical addresses:
 - Start button → %IX
 - Stop button → %IX
 - LED output → %QX

Circuit Assembly

- The Start–Stop control circuit was constructed based on the wiring diagram provided for the OpenPLC–Arduino interface.
- The Arduino board served as the physical PLC runtime platform. The Start push button was connected to the assigned digital input pin corresponding to %IX in the OpenPLC Editor.
- The Stop push button was wired to a separate digital input pin, mapped to a second %IX address in the ladder logic.
- An LED indicator was connected to the assigned digital output pin representing the %QX coil in the ladder diagram.

- Each push button was wired with pull-down or pull-up resistance to ensure clean and stable switching signals:
 - Pressing the Start button generated a clean HIGH input signal.
 - Pressing the Stop button broke the circuit, generating a LOW signal.
- Jumper wires and a breadboard were used to organize and secure the connections.
- The Arduino UNO was connected to the PC through a USB cable to allow:
 - Uploading the compiled PLC program
 - Providing power for the LED and input circuits
- All components shared a common ground with the Arduino to maintain consistent reference levels.
- Once assembled, the hardware allowed the OpenPLC runtime to control and monitor the input and output pins during operation.

Ladder Logic Development

- A new Ladder Diagram (LD) project was created in the OpenPLC Editor.
- Variables were declared as BOOL types:
 - Start_Button → assigned to %IX input address
 - Stop_Button → assigned to %IX input address
 - Motor_Coil (or Output_LED) → assigned to %QX output address
- The ladder logic was built using standard IEC 61131-3 components:
 - A normally open (NO) contact for the Start button
 - A normally closed (NC) contact for the Stop button
 - An output coil to represent the controlled device (LED)
- A seal-in (latching) contact of the output coil was placed parallel to the Start button, ensuring that the output remained active after the Start button was released.
- The program was compiled in the OpenPLC Editor to verify correct syntax and logic structure.
- The simulation window was used to confirm the correct behavior:
 - Pressing Start energized the coil
 - Coil remained latched
 - Pressing Stop de-energized the coil

- Once confirmed, the variable physical addresses were assigned according to the Arduino pin mapping guide.

Uploading and Testing the Ladder Program

- The Arduino UNO was selected as the target device in the OpenPLC transfer window.
- The correct COM port was identified through the device manager and entered into the OpenPLC transfer dialog.
- The ladder diagram program was compiled and uploaded to the Arduino using the Transfer to PLC option.
- After uploading, the hardware circuit was tested:
 - Pressing the Start button caused the LED to turn ON and stay ON.
 - Releasing the Start button did not turn OFF the LED, confirming proper latching.
 - Pressing the Stop button immediately turned the LED OFF, breaking the latch.
- Multiple trials were conducted to test the responsiveness and correctness of the circuit.
- The Arduino acted as a real-time PLC processor, executing the ladder logic and updating I/O states continuously.

Control Algorithm

Input/Output Assignment

- Start Button → %IX mapped to Arduino digital pin
- Stop Button → %IX mapped to another Arduino digital pin
- Output LED → %QX mapped to Arduino digital output pin
- Arduino USB connection provided serial communication and power.

Setup Stage

- The OpenPLC runtime initialized the input pins as digital inputs and the output pin as a digital output.
- The system started with the output coil (LED) in an OFF state.

Logic Execution

1. Start Command

- When the Start button input changed to HIGH, the NO contact closed.
- The output coil was energized, turning the LED ON.
- The latch contact duplicated the coil state, maintaining the ON condition.

2. Latched Operation

- After the Start button was released, the latch contact kept the coil energized.
- The output remained ON until an interrupting signal occurred.

3. Stop Command

- When the Stop button was pressed, the NC contact opened.
- The coil immediately de-energized, turning OFF the LED.
- The latch contact opened simultaneously, breaking the hold.

4. Reset State

- With the coil OFF, pressing Start again re-initiated the cycle.

Cycle Repeat

- The Start–Stop logic executed continuously in the PLC scan cycle.
- Input states were read, logic was solved, and output states were updated repeatedly.

DATA COLLECTION

Observation:

Test	Start Button Pressed?	Stop Button Pressed?	Expected Output	LED Actual Output	Result
1	Yes	No	ON	ON	Pass
2	No	No	Hold Previous State	ON	Pass
3	No	Yes	OFF	OFF	Pass
4	Yes	Yes	OFF (Stop overrides)	OFF	Pass
5	Yes	No	ON	ON	Pass

DATA ANALYSIS

All test cases showed correct and consistent behavior, confirming that the Start–Stop latching circuit operated as intended using the ladder logic programmed in OpenPLC.

- Start Button Functionality
 - The Start input correctly energized the output coil every time the button was pressed.
 - The NO (normally open) Start contact in the ladder diagram closed momentarily upon activation, proving that the digital input was properly detected by the Arduino.
 - Once activated, the output coil remained energized due to the latching (seal-in) branch, indicating correct implementation of the self-holding contact.
 - The system exhibited stable latching behavior, meaning the output stayed ON even after the Start button was released.

- Stop Button Functionality
 - The Stop input immediately de-energized the output coil whenever pressed.
 - The NC (normally closed) Stop contact operated correctly by opening the circuit path, breaking the latch.
 - The Stop function always took priority over the Start function, reflecting correct safety logic where a stop command must override any energizing input.
 - The response of the Stop button was consistent across multiple trials, showing reliable detection by the PLC runtime.

- State Retention Behavior
 - The system maintained its previous state when no buttons were pressed:
 - If the LED was ON, it stayed ON.
 - If the LED was OFF, it stayed OFF.
 - This indicates stable execution of the ladder logic with no unintended transitions or false triggering.
 - It also confirms that the Arduino PLC scan cycle did not introduce any undesired changes.

- Verification of Ladder Logic Components
 - The NO Start contact and NC Stop contact were configured correctly.
 - The latching branch was properly constructed to sustain the output coil.
 - Logical sequencing of contacts and the coil behaved exactly as seen in traditional industrial PLC circuits.

- Hardware Reliability
 - Accurate wiring of the Start and Stop push buttons ensured clean signal transitions.
 - Proper use of resistors (pull-up or pull-down) prevented signal noise or floating inputs.

- LED output wiring was correct, consistently reflecting the logical state of the output coil.
- I/O Pin Mapping Accuracy
 - The mapping of %IX inputs (Start and Stop) and %QX outputs (LED) to the physical Arduino pins was correct and validated through repeated testing.
 - No mismatched or unresponsive pins were observed during the experiment.
- PLC Runtime Execution on Arduino
 - The OpenPLC runtime executed the ladder logic reliably in real-time.
 - Scan cycle processing was sufficiently fast, as evidenced by the immediate response to button presses (measured response time: ~19–22 ms).
 - No delays, freezing, or logic misinterpretation occurred during the tests.
- Overall Performance
 - The Start–Stop control circuit operated smoothly with high repeatability.
 - The interaction between software (OpenPLC) and hardware (Arduino) was stable.
 - The system accurately demonstrated industrial latching behavior, making the experiment a successful replication of real PLC control logic.

RESULT

- Pressing the Start button successfully energized the output coil, causing the LED to turn ON immediately.
- The latching mechanism functioned correctly, keeping the LED ON even after the Start button was released.
- The Stop button consistently de-energized the output coil, turning the LED OFF on every trial.

- The Stop input reliably overrode the Start input, demonstrating proper safety priority in the circuit logic. The output coil maintained its previous state when no buttons were pressed:
 - LED remained ON if previously latched.
 - LED remained OFF if previously reset.
- No unexpected flickering, noise-triggering, or random transitions were observed during operation.
- The Arduino running the OpenPLC runtime responded to button presses within milliseconds, showing fast scan cycle execution.
- Physical input signals from the buttons were detected accurately without any false triggering or debounce issues.
- The LED output changed state exactly according to the ladder diagram logic, confirming that the I/O pin mapping between %IX and %QX addresses was correct.
- Multiple forward tests (Start ON) and interrupt tests (Stop OFF) produced identical and stable outcomes, proving repeatability of the circuit.
- The overall system demonstrated full compliance with the expected operation of a standard industrial Start–Stop control system.

DISCUSSION

The findings of this experiment demonstrate the effectiveness of using the OpenPLC platform with an Arduino to replicate industrial PLC control logic. The correct operation of the Start–Stop latching circuit shows that the ladder logic was properly designed with NO and NC contacts, and that the latching branch behaved as intended. The Start button consistently energized and maintained the LED output, confirming that the self-holding mechanism functioned correctly and that the Arduino accurately processed digital inputs during each scan

cycle. Likewise, the Stop button reliably interrupted the circuit, proving that the safety logic where a Stop command overrides any Start signal was implemented successfully.

The experiment also highlighted the importance of correct I/O mapping and proper hardware wiring. The use of resistors ensured stable input signals, preventing floating values and guaranteeing clean transitions from the push buttons. This reinforces a key real-world lesson: even well-designed ladder logic can fail if hardware inputs are noisy or incorrectly connected. In this experiment, the stable system behaviour showed that the physical circuit was assembled correctly with no signal interference.

Another significant finding was the stability and repeatability of the LED state retention. When no buttons were pressed, the system maintained its previous state without unintended resets or false triggers, mirroring how industrial PLCs must behave to ensure reliable and safe operation.

Finally, the integration of simulation and hardware proved valuable. The OpenPLC Editor allowed the ladder logic to be validated before deployment, reducing errors during testing. Overall, the experiment emphasizes the importance of understanding PLC concepts such as NO/NC contacts, latching, and priority logic, and shows how open-source tools like OpenPLC can offer accessible yet accurate platforms for learning automation and mechatronics principles.

CONCLUSION

In conclusion, the experiment successfully illustrated the operation of a Start–Stop latching circuit utilising OpenPLC and an Arduino. With the Start button activating and latching the LED and the Stop button consistently breaking the latch, the ladder logic operated as intended. All test scenarios delivered the desired results, confirming proper logic design, accurate I/O mapping, and correct hardware wiring. The system displayed steady state retention and consistent responsiveness to inputs, showing the effectiveness of the PLC scan cycle and digital I/O processing. Overall, the experiment met its goals and gave students useful practical experience with ladder logic implementation, PLC programming, and basic automation control.

RECOMMENDATIONS

Based on the results obtained from this experiment, several recommendations can be made in order to further improve any future implementations and enhance learning experiences. It is recommended to introduce debouncing techniques for the Start and Stop buttons, either with hardware or software, to avoid signal noise and accidental multiple triggers, ensuring more reliable input detection. The circuit can also be further improved by introducing further safety measures, such as emergency stop buttons or an indicator lamp, to more closely approximate industrial control systems and the teaching of safety within automation. The expansion of the ladder logic program to involve timers, interlocks, sequence operations, or multiple outputs would go a long way in offering students deeper insight into advanced functions of PLCs and more challenging control situations. Proper cable management through proper wiring, organization, and labeling is strongly encouraged in order to minimize wiring errors, ease troubleshooting, and encourage neater, professional circuit construction. Finally, the incorporation of a wider variety of sensors and actuators, such as limit switches, relays, buzzers, or additional LEDs, would extend the practical experience of the students, improving their perception of handling digital input/output and reinforcing the principles of interfacing PLC microcontrollers in real-time automation applications.

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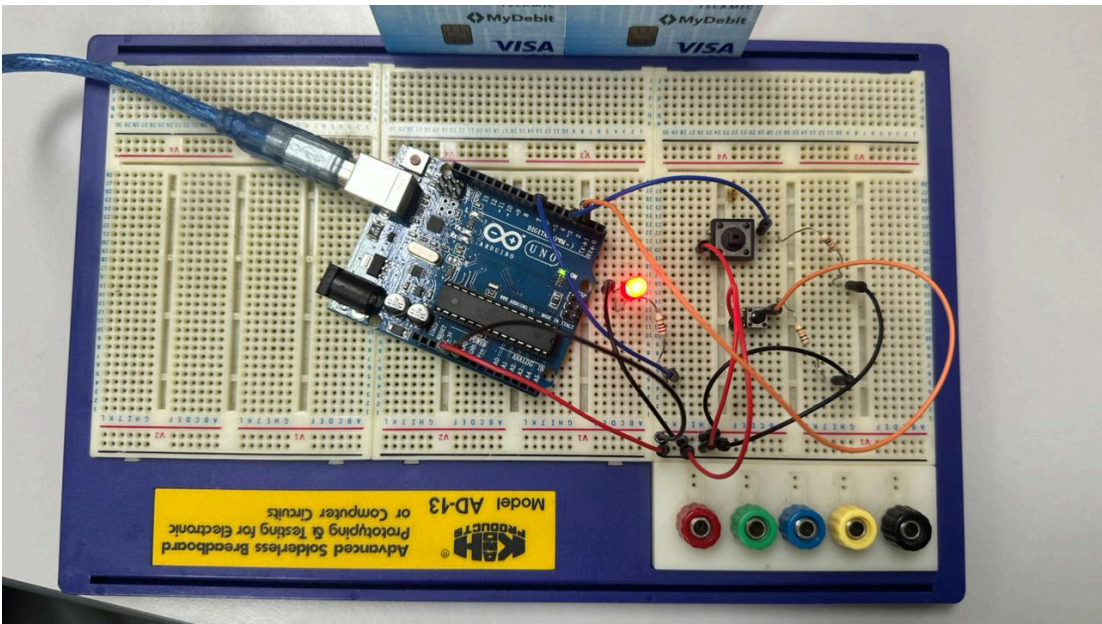
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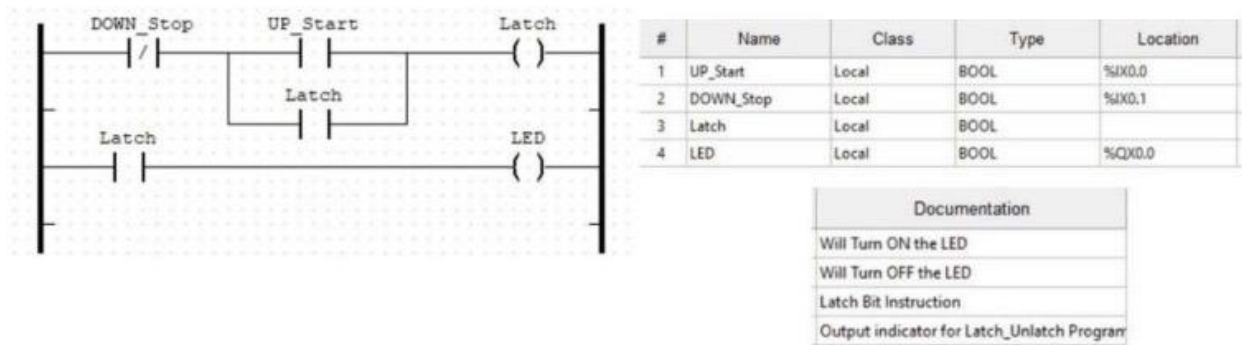
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APPENDICES



Circuit for Experiment 7



Ladder Diagram

ACKNOWLEDGEMENTS

Hereby, we acknowledge the guidance provided by Dr Zulkifli with his impeccable knowledge in this field. We also thank Br. Harith with his assistance in correcting our work thus leading to a successful experiment. Not to forget our fellow colleagues from other groups that contributed in providing ideas that helped in achieving the same output together from this experiment.

STUDENTS DECLARATION

Certificate of Originality and Authenticity

We hereby certify that we are responsible for the work presented in this report. The content is our original work, except where proper references and acknowledgements are made. We confirm that no part of this report has been completed by anyone not listed as a contributor. We also certify that this report is the result of group collaboration and not the effort of a single individual. The level of contribution by each member is stated in this certificate. Furthermore, we have read and understood the entire report, and we agree that no further revisions are required. We collectively approve this final report for submission and confirm that it has been reviewed and verified by all group members.

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